

Enriching Existing Educational Video Datasets to Improve Slide Classification and Analysis

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ABSTRACT

Educational content is increasingly available online. The most common media types for such content are audio and video. Many of these educational video are unprocessed – they are simply captured with a camera and then uploaded onto video servers for viewers to watch. We believe that automatic video analysis that recovers the structure of educational videos would allow learners and creators to fully exploit the semantics of the captured content. In this paper, we outline our research plan towards better semantic understanding of educational videos. We highlight our work done so far: (i) we extended the FitVid Dataset to exploit the semantics of any type of learning video; and (ii) we improved upon existing state-of-the-art slide classification techniques.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; • **Computing methodologies** → **Computer vision**.

KEYWORDS

datasets, neural networks, multimodal analysis, educational video content, summarization, object detection, slide classification

ACM Reference Format:

Travis Seng. 2022. Enriching Existing Educational Video Datasets to Improve Slide Classification and Analysis. In *Proceedings of the 30th ACM International Conference on Multimedia (MM '22)*, October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3503161.3548758>

1 MOTIVATION AND NOVELTY

The amount of online learning content keeps growing every day and many universities, teachers, and researchers are making these resources available to everyone publicly and easily, especially video content. However, many of these videos are just raw or lightly edited recordings of a presentation (lecture class, computer screen sharing) making it difficult to quickly grasp its content or reusing it.

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MM '22, October 10–14, 2022, Lisboa, Portugal

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ACM ISBN 978-1-4503-9203-7/22/10...\$15.00

<https://doi.org/10.1145/3503161.3548758>

A lot of work has been done toward the goal of making it easier to understand these videos such as video summarization [13, 15, 17], structure understanding [1, 9, 12, 14, 19], content extraction [1, 2, 5, 6, 8, 9, 12, 14, 19] and better browsing tools [1, 9, 12, 14, 19]. An aspect that has been neglected so far is the goal of easing the task of the video's creator. Proposing a tool that allows users to easily edit their video, add more slides, pictures, change its structure and record additional content on the fly would facilitate updating these videos.

It is also worth noting that previous work only focus on a limited number of shot types, mainly clean slides, thus limiting the number of videos that can take advantage of such work. Presentation videos have a lot of different shot types, as shown in Figure 1. Moreover, the COVID-19 pandemic put online classes and videoconferences to the spotlight, effectively increasing the variety of video (Type C). As it stands, there is no method that performs well on all video types. One goal of my thesis is to design a system that extracts the semantics of any type of presentation recording based on all available modalities (video, audio and more) for several applications including summarization, quick editing and content adaptation.

This paper focuses on the work that has been done so far, i.e. the visual content analysis of any type of presentation videos. The key points of the work are to propose a new dataset, to look into the potential applications and to discuss future work.

2 BACKGROUND AND RELATED WORK

2.1 Educational video analysis and summarization

In [14], the authors capture the structure of a presentation by synchronizing a direct static camera feed with given slides and output a structured document where users could easily navigate through. Many contributions have since iterated on it such as using the audio transcription and OCR to generate a better automatic table of content and new interfaces for navigation and browsing [2, 9, 12, 19]. In [8], the authors designed a system that understands the elements of a slide to dynamically modify its formatting for better comfort and readability on mobile devices. Lecture videos summarization comes in many forms: the generation of a table of content is a form of summarization, but [17] summarizes the audio transcription and [15] generates lecture notes from the content of the slides. Video skimming and key frames generation [7] are other ways to achieve this. The main problem is that most of these approaches only perform with a computer screen feed (Types A and C).

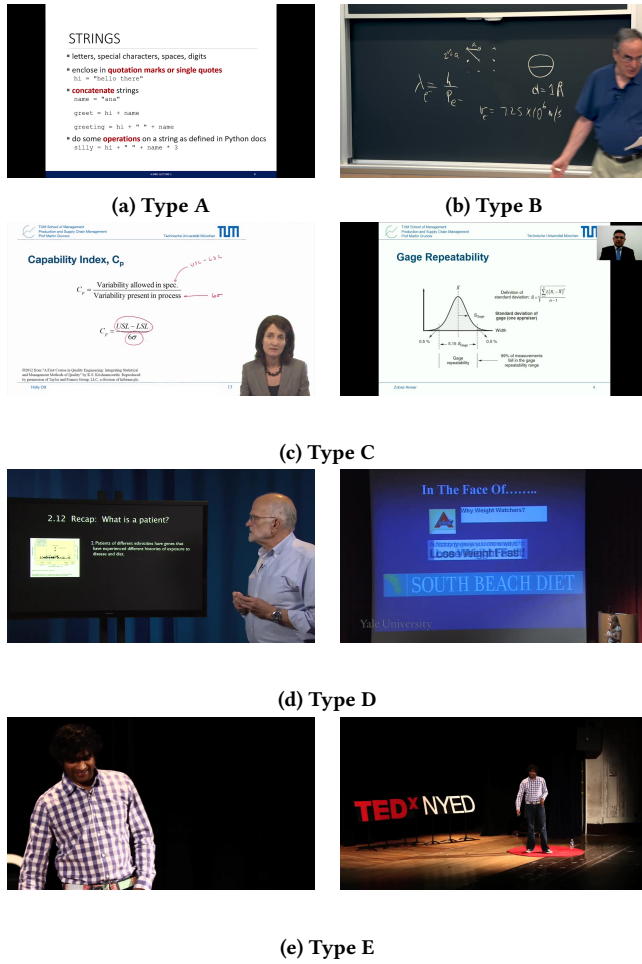


Figure 1: Five types of shots in our research. A video can switch back and forth between every type, shows the presenter or not, and can have a moving or static camera. (a) Type A shows a computer slide screen. (b) Type B shows a direct camera feed with the presenter writing on a whiteboard. (c) Type C presents a speaker and computer screen overlaid together. (d) Type D shows a direct camera feed with a presenter and the projected screen. (e) Type E shows only the presenter or the audience. It does not contain direct information such as slides, computer screen or hand writings.

2.2 Existing datasets

Previous work has shown that presentation videos can be classified into six types of video [11]. Here, we prefer to list the types of shot that appear in these videos for our analysis in Figure 1.

Some annotated datasets have been published. Table 1 describes existing annotated presentation video datasets. Most of these are not suitable for our problem for multiple reasons: small size, lack of diversity of subjects, not enough types of shots, different annotations. To solve our problem, we need a large and high quality dataset designed for semantic units (titles, text boxes, pictures, diagrams, hand writings, etc.) that contains all types of shots and a large

amount of diverse subjects (e.g. more than just science courses). FitVid Dataset [8] seems to be a good fit but it contains only clean slides (Type A and C).

3 APPROACHES AND RESULTS

In this section, we discuss the work that has been done so far toward the goal of extracting the visual content of a given video.

It is important to first determine what should be extracted. Obviously, everything shown or written by the presenters is relevant, but we could also extract the presenters themselves to detect their gestures (for example, if they point at something on the slide).

During a presentation, the slides usually hold most of the information and their content needs to be extracted and recognized.

3.1 Slide content extraction and recognition

As pointed out by the related work, there are only few high quality datasets for content detection, and they do not contain all types of videos. For this reason, we propose a new dataset that has every type of content and type of editing and filming.

We use the FitVid Dataset [8] as the training dataset and train a YOLOX [4] object detection network. The dataset contains the following labels: title, text box, picture, chart, figure, diagram, table, schematic diagram, header, footer, hand writing, and instructor. This network is then used to generate labels on diverse images from the existing datasets and free lecture videos on YouTube. The generated labels are then manually corrected and added to the training dataset. We iterate this step multiple times until we obtain a good amount of labeled new images.

Around 2000 images have been added to the FitVid Dataset through this method, and we will continue to add more. Figure 2 illustrates how the enriched dataset allows getting better results. Some detection results on every type of shots are shown in figure 3.

Training only on the original FitVid Dataset gives 59% precision, 44% recall and 50% mAP@0.5 on the test images, while training on the enriched dataset gives **76% precision, 87% recall and 84% mAP@0.5** (all labels).

This dataset is intended to be released publicly.

3.2 Shot classification

Different sequence types can appear in a presentation. The presenters may show something to the public, write on a whiteboard, show a video or their slides. This knowledge is useful to perform further tasks, such as whiteboard summarization [3, 10].

We want to compare our results with the last known contribution on shot classification, which is Zhao et al. work [18] to improve upon the state of the art. We follow Zhao's method for generating the shots (also referred to as key frames) for 18 test videos given by the authors. The goal is to detect whether a shot is "slide" shot (shot containing at least a readable projected screen or displaying a computer screen) or "non-slide" shot (everything else). Our content detection allows us to classify a shot with heuristics based on the detection results: size, location and numbers and type of objects are used to determine whether a shot is a "slide" shot or a "non-slide" shot. For example, if our system detects multiple text boxes and a

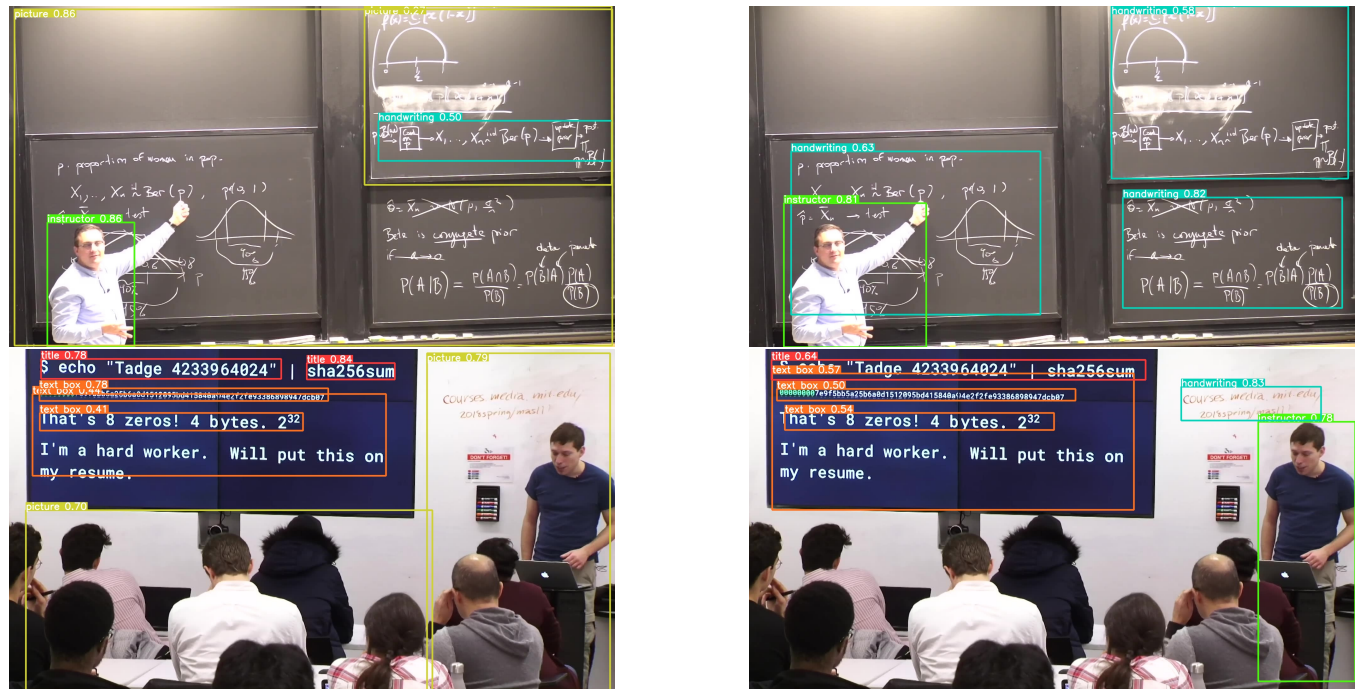


Figure 2: Object detection results. On the left: YOLOX trained on original FitVid. On the right: YOLOX trained on new extended dataset. The old dataset didn't have enough diversity to understand direct camera input. The majority of the "picture" false positives are corrected with the extended dataset.



Figure 3: Different results on different types of shot from our test enriched dataset. Our method successfully captures pictures, hand writings, presenters and text boxes in every type of shot.

Dataset name	Type of shot	Size	Annotation type
Wise [6]	projected slides with and without presenter (B, D, E)	1300 images from the ClassX dataset	14 text, 6 image-based and 4 structural classes
Spase [5]	clean slides with talking head (A, C)	2000 complex slides from Slideshare-1M	14 text, 6 image-based and 4 structural classes
FitVid [8]	clean slides with talking head (A, C)	5,527 video frames sampled from 66 courses	12 classes, box labels (object detection)
Lecture2Notes ¹	projected slides with and without presenter, clean slides, audience, whiteboard (A, B, D, E)	5502 slide shots, 5040 presenter and slide shots and 5057 frames other shots	Classification (slide, presenter, whiteboard, other)

Table 1: Existing datasets description

	Talkminer [1]	Zhao et al [18]	Ours
Precision	81,6	92,6	97,7
Recall	74,7	86,3	97,8
F1	74,2	88,6	97,7

Table 2: Performance of different methods on shot classification. Average values over 18 videos from [18] evaluation dataset

Video	Audio	Metadata
screen sharing, slides, camera (audience, presenter or something else), talking head, whiteboard	different voices, transcription	additional material, pointer trajectory, annotations, commentaries, danmaku [16], chat

Table 3: The different modalities of a learning video content

title, the shot will be classified as a "slide" shot. The Table 2 shows that our method has the best performance.

4 ON-GOING AND FUTURE WORK

Even though we have good results, there is still mislabeling between text boxes, titles, hand writings and some texts are not correctly detected. We are investigating how **hierarchical object detection and temporal information** can help increase the accuracy of the classification.

This paper only focuses on the visual aspect, but the longer-term goal of the thesis is to take every modality into account. Table 3 shows what kind of modalities we can face and we can use to improve semantic understanding. For example, audio will be used for transcription and detection of changes of speakers, which simplifies the structuring and automatic indexing of the content. Moreover, secondary content (e.g., presentations shared on screen and provided as PDFs, textual streams conversations, commentaries) can help to specialize the structuring algorithm. Even pointer trajectories [19] and presenters gestures can be used to highlight specific content.

¹Dataset available on <https://lecture2notes.readthedocs.io/en/latest/>

The work presented here is part of the first axis of my thesis which is the structuring, restitution and indexing of a raw educational content. The goal is to design a system that automatically render a presentation video into an interactive, structured and more readable document. The first step is to detect the relevant visual content and separate it from the background. Then, after semantic analysis by classification and text analysis, we can identify the different elements of the video and understand the structure of the document. The evaluation of such method will be quantitative (i.e. did we capture all the information) and qualitative. User studies will be conducted to evaluate the quality of the restitution. For example, does the new representation lead to a better and faster understanding of the content?

The second axis is not a scientific priority. It will consist in designing protocols for editing and transforming video data. This is the tool we want to create to allow users to easily edit their video. For example, users will be able to resize pictures, move the text and change the order of the slides on raw recordings. Such tools already exist and we are currently investigating how to integrate our methods.

The last axis is to optimize the new representation, i.e. restore a pedagogical content or a meeting report by optimizing certain properties (compactness, intelligibility, aesthetics, etc.). The central point of the work to be done lies in the evaluation of the expected properties. The identification of metrics correlated to the subjective evaluations that user studies will provide is envisaged.

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